Experiment:9 Computer-Aided Design and Optimization of a Microwave Impedance-Matching Network and Frequency-Selective Filter

Aim: To simulate microwave semiconductor devices using MATLAB and analyze the frequencyconversion characteristics of a microwave mixer.

Apparatus:

Computer/Laptop

Matlab Software

Procedure:

- Open MATLAB and create a new script.
- Initialize the characteristic impedance, load impedance, design frequency, and frequency range.
- Calculate the reflection coefficient and return loss of the microwave load.
- Model the impedance-matching network and calculate the input impedance and S11 parameter.
- Design a band-pass filter with a centre frequency of 2 GHz and bandwidth of 0.5 GHz.
- Plot the S11 response and filter magnitude response against frequency.
- Run the program and observe the graphs and calculated values in the Command Window.
- Analyze the results to understand impedance matching and frequency-selective filter characteristics.

Code:

```
clc;

clear;

close all;

Z0 = 50;

ZL = 75 + 1j*25;

f0 = 2e9;

c = 3e8;

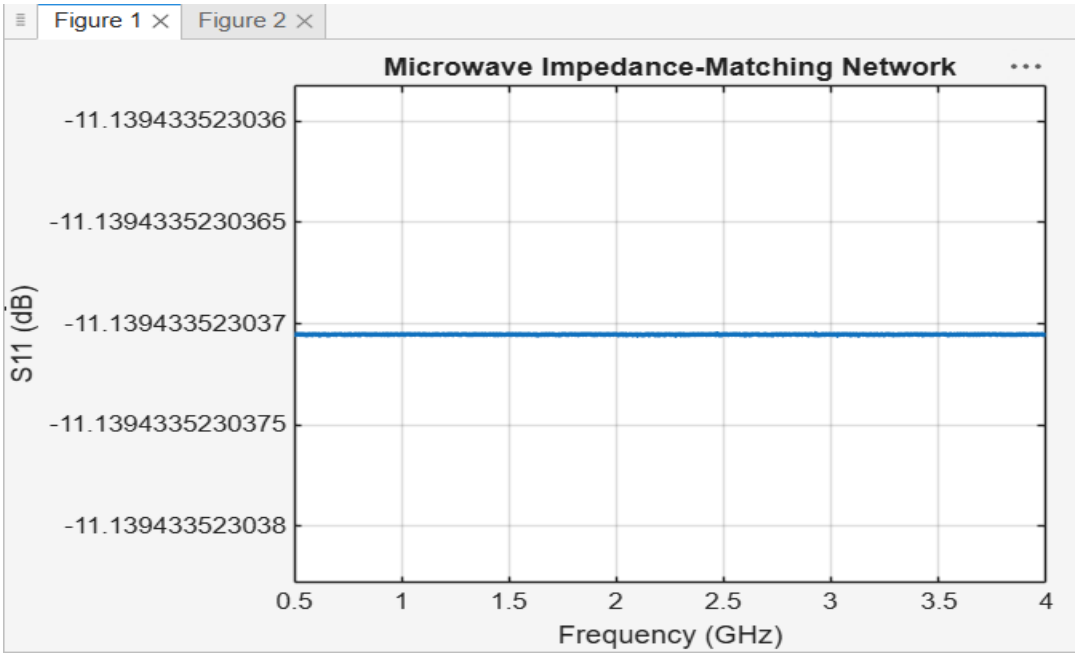
f = linspace(0.5e9, 4e9, 1000);

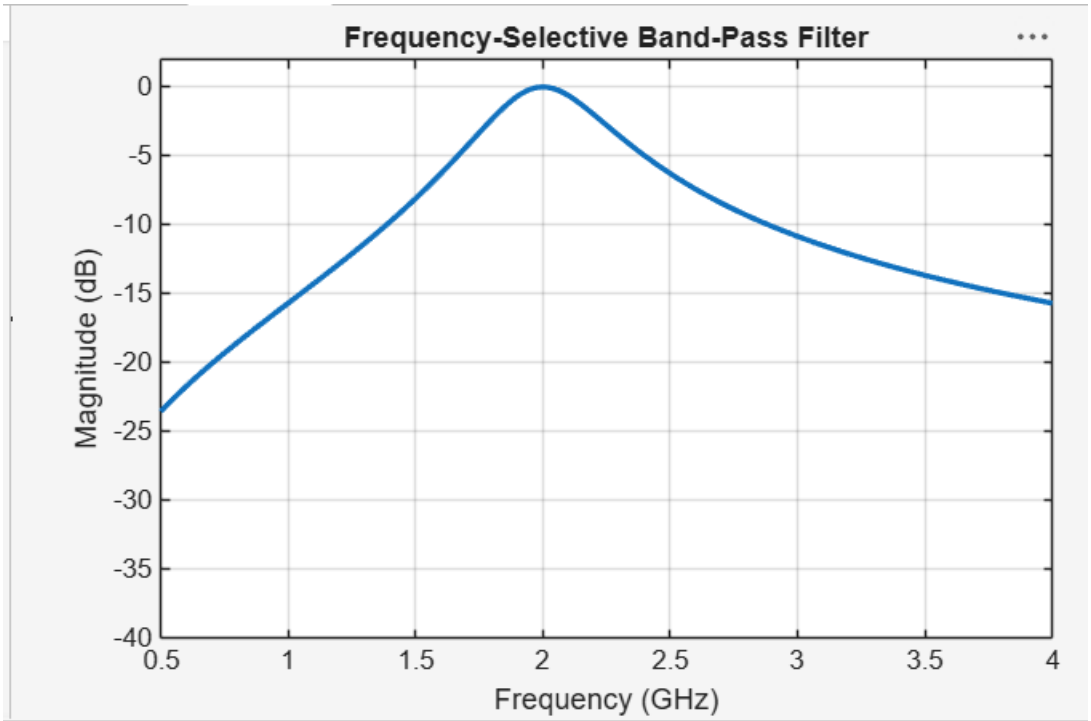
lambda = c/f0;
```

```
GammaL = (ZL-Z0)/(ZL+Z0);
RL = -20*log10(abs(GammaL));
theta = linspace(0, 2*pi, 1000);
Gamma = GammaL*exp(-1j*theta);
fprintf('Characteristic Impedance = %d Ohms\n', Z0);
fprintf('Load Impedance = %.1f + j%.1f Ohms\n', real(ZL), imag(ZL));
fprintf('Reflection Coefficient Magnitude = %.4f\n', abs(GammaL));
fprintf('Return Loss = %.2f dB\n', RL);
fprintf('Design Frequency = %.2f GHz\n', f0/1e9);
fprintf('Wavelength = %.4f m\n', lambda);
% Impedance matching network
Zin = Z0*(ZL + 1j*Z0*tan(2*pi*f/f0*0.25)) ./ ...
    (Z0 + 1j*ZL*tan(2*pi*f/f0*0.25));
GammaIn = (Zin-Z0)./(Zin+Z0);
S11 = 20*log10(abs(GammaIn)+1e-12);
figure;
plot(f/1e9, S11, 'LineWidth', 2);
grid on;
xlabel('Frequency (GHz)');
ylabel('S11 (dB)');
title('Microwave Impedance-Matching Network');
% Band-pass filter design
fc = 2e9;
BW = 0.5e9;
Q = fc/BW;
x = f/fc;
```

```
H = 1./sqrt(1 + Q^2*(x - 1./x).^2);
H_dB = 20*log10(H);
figure;
plot(f/1e9, H_dB, 'LineWidth', 2);
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Frequency-Selective Band-Pass Filter');
ylim([-40 2]);
% Filter response and bandwidth
idx = find(H_dB >= -3);
if ~isempty(idx)
    fprintf('Approximate Filter Bandwidth = %.3f GHz\n', ...
        (f(idx(end))-f(idx(1)))/1e9);
end
```

Output:





Result:

The microwave impedance-matching network and frequency-selective band-pass filter are simulated using MATLAB. The plots illustrate the reflection characteristics and frequency-selective behavior of the microwave circuit.